

Nucleophilic Cyclopropanation Reaction with Bis(iodozincio)methane by 1,4-Addition to α,β -Unsaturated Carbonyl Compounds

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Abstract: Treatment of α,β -unsaturated ketones with an electrophilic site at the γ -position in the presence of trimethylsilyl cyanide with bis(iodozincio)methane afforded the (*Z*)-silyl enol ether of the β -cyclopropyl substituted ketone in good yields. The reaction proceeds by 1,4-addition to form an enolate, and its sequential intramolecular nucleophilic

attack to an adjacent electrophilic site. The reaction of γ -ethoxycarbonyl- α,β -unsaturated ketone and bis(iodozincio)methane in the presence of

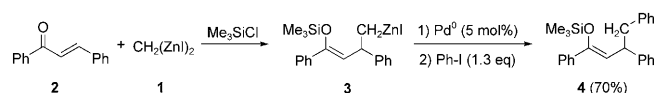
Keywords: cyclopropanation • diastereoselectivity • enols • nucleophilic addition • silylation

trimethylsilyl cyanide afforded 1-ethoxy-1-trimethylsilyloxycyclopropane derivatives, which can be regarded as the homoenolate equivalent. Additionally, reaction of the obtained homoenolate equivalents with imines give 1-(*E*)-alkenyl-2-(1-aminoalkyl)alkanols diastereoselectively.

Introduction

Many examples of 1,4-addition of an organozinc reagent to α,β -unsaturated ketones have been reported.^[1] We had also reported that the *gem*-dizinc reagent undergoes 1,4-addition to α,β -unsaturated ketones.^[2] For example, treatment of chalcone (**2**) with bis(iodozincio)methane (**1**) in the presence of chlorotrimethylsilane afforded (*Z*)-trimethylsilyl enol ethers of β -iodozinciomethylketone **3** by 1,4-addition (Scheme 1).^[3] The sequential coupling reaction at the introduced zinciomethyl group with other electrophiles, such as phenyl iodide, was also possible in the presence of Pd⁰.

Instead of performing the intermolecular coupling shown in Scheme 1, an intramolecular process would afford cyclic



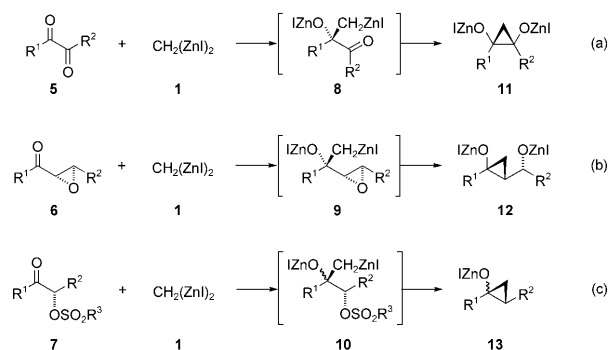
Scheme 1. 1,4-Addition of bis(iodozincio)methane (**1**) to chalcone (**2**) and sequential coupling with phenyl iodide in the presence of Pd⁰.

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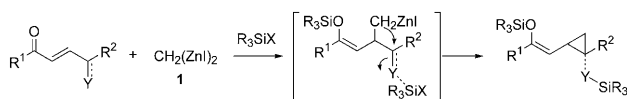
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compounds. We have already shown that a sequential nucleophilic reaction is an efficient route for cyclopropane ring formation.^[4] When 1,2-diketone **5**, α,β -epoxyketone **6**, or α -sulfonyloxyketone **7** was treated with **1**, 1,2-addition to the carbonyl group proceeds to generate organozinc species, **8**, **9**, or **10**, respectively. In this case, a sequential intramolecular nucleophilic attack of the carbon–zinc bond to another carbonyl group or the leaving group, epoxy group, or sulfonyloxy group, proceeds readily, and the corresponding zinc alkoxide of cyclopropanol **11**, **12**, or **13** was obtained (Scheme 2).

In this context, we planned a sequential process of 1,4-addition and cyclopropane formation using *gem*-dizinc **1** to produce a β -cyclopropyl substituted silyl enol ether. As



Scheme 2. Nucleophilic cyclopropanation of bis(iodozincio)methane (**1**) with 1,2-diketone (**5**), α,β -epoxyketone (**6**) or α -sulfonyloxyketone (**7**) by iodozinciomethylation.



Scheme 3. Nucleophilic cyclopropanation of bis(iodozincio)methane (**1**) by 1,4-addition.



Scheme 4. Cyclopropanation of γ -acetoxy- α,β -unsaturated ketone **14a** with bis(iodozincio)methane (**1**).

shown in Scheme 3, we examined the treatment of a α,β -unsaturated ketone carrying an electrophilic site at the γ -position with bis(iodozincio)-methane (**1**) in the presence of a silylation reagent (R_3SiX).

Results and Discussion

A mixture of γ -acetoxy- α,β -unsaturated ketone **14a** (1.0 mmol), triethylamine (2.5 mmol), and trimethylsilyl cyanide (2.2 mmol) in THF was treated with **1** (1.2 mmol) at 25 °C for 5 min. After aqueous work-up, (*Z*)-silyl enol ether of β -cyclopropyl ketone **15a** was obtained in 83% yield stereoselectively (Scheme 4).^[2,5]

Other examples are shown in Table 1. In all cases, the products were obtained as (*Z*)-form enolates selectively. When γ -acetoxy- γ -methyl- α,β -enone **14e** was treated with **1** in the presence of trimethylsilyl cyanide, (*Z*)-enolate **15e** was obtained in 80% yield with low diaste-

Table 1. Cyclopropanation of γ -acyloxy- α,β -unsaturated ketone **14** with bis(iodozincio)methane (**1**) by 1,4-addition.

entry	substrate	conditions	product	yield [%] ^[a]	d.r. ^[b]
1		25 °C, 3 h		86	–
2		25 °C, 3 h		83	–
3		25 °C, 3 h		51 ^[c]	–
4		0 to 25 °C, 12 h		80	59:41 ^[d]
5	14e	–40 to 25 °C, 12 h	15e	80	60:40
6		0 to 25 °C, 12 h	15e	75	66:34
7		0 to 25 °C, 12 h		82	79:21 ^[e]
8		0 to 25 °C, 12 h		27 ^[f]	85:15

[a] Isolated yields. [b] The diastereomeric ratios were determined by ¹H NMR. [c] **15d** was obtained as equimolar mixture of (*E*)- and (*Z*)-isomers after purification. [d] The stereochemistry of substituents on cyclopropane ring in major isomer was *trans*. See Supporting Information. [e] The diastereomeric ratio was determined after converting to the corresponding ketone with 1 N aqueous HCl. [f] The starting material was recovered in 65% yield.

Abstract in Japanese:

シアン化トリメチル共存下、 γ 位に求電子部位を有する α,β -不飽和ケトンに対してビス(ヨードジンシオ)メタンを作用させると、 β 位にシクロプロピル基を有するケトンの(*Z*)-シリルエノールエーテルが高収率で得られた。反応は共役付加によってエノラートを生成し、続いて隣接する求電子部位に対して分子内求核反応を経ることによって進行している。 γ 位にエトキシカルボニルを有する α,β -不飽和ケトンと二亜鉛種との反応では1-エトキシ-1-トリメチルシロキシシクロプロパン誘導体が得られるが、これはホモエノラート等価体とみなすことができる。我々はこのホモエノラート等価体とイミンとの反応において1-(*E*)-アルケニル-2-(1-アミノアルキル)アルカノールが高ジアステレオ選択的に得られることも合わせて見いだした。

reoselectivity with respect to the stereochemistry of cyclopropane ring (entry 4). Starting the reaction at –40 °C did not improve the diastereomeric ratio (entry 5). In these cases, the diastereofacial selectivity of 1,4-addition to α,β -enone will determine the diastereomeric ratio of the product, because the subsequent intramolecular nucleophilic substitution of the remaining carbanion site to the adjacent acyloxy substituted carbon should be stereospecific. Yamamoto has reported the diastereofacial selectivity of 1,4-addition of carbon nucleophiles to α,β -unsaturated carbonyl derivatives.^[6] In general, the diastereoselectivity can be explained by the modified Felkin–Anh model when 1,4-additions proceed through the Bürgi–Dunitz trajectory^[7] (Figure 1). The model could be applied to our reaction, so the size of the substituents at the γ -position was examined to determine the diastereoselectivity. The reaction of γ -(2,4,6-trimethyl-

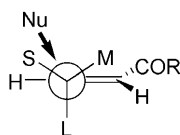


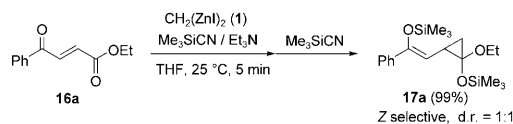
Figure 1. Modified Felkin–Anh Model by Yamamoto.

benzoyloxy)- α,β -enone **14f** improved the diastereomeric ratio only slightly (entry 6). In the case of γ -isopropyl-substituted enone **14h**, the diastereomeric ratio was improved to 85:15, but the reaction became sluggish (entry 8).

Instead of **14**, γ -ethoxycarbonyl- α,β -enone **16** was also examined as a substrate. A mixture of ethyl (*E*)-3-benzoylacrylate (**16a**, 1.0 mmol), triethylamine (2.5 mmol), and trimethylsilyl cyanide (1.1 mmol) in THF was treated with **1** (1.2 mmol) at 25°C for 5 min. Additional trimethylsilyl cyanide (1.1 mmol) was added and the mixture was stirred for 10 min at room temperature. After aqueous work-up, 1-ethoxy-1-trimethylsiloxycyclopropane **17a** was obtained in 99% yield (Scheme 5). The configuration of the enolate was the (*Z*)-form without any diastereoselectivity from the cyclopropane formation reaction.

Other examples are shown in Table 2. In all cases, the products **17** were obtained as an equimolar mixture of diastereomers, as the intramolecular cyclopropanation proceeded without diastereoselectivity (from **18** to **17** in Scheme 6). When chlorodimethylphenylsilane was used as silylation reagent instead of trimethylsilyl cyanide, the corresponding phenyldimethylsilyl ether derivatives **17a'–d'** were obtained (entries 5–8).

The product **17** has a 1-ethoxy-1-siloxycyclopropane moiety, which is known to form an ester homoenolate by cyclopropane ring opening. Nakamura and Kuwajima had reported that treatment of 1-siloxy-1-alkoxycyclopropanes



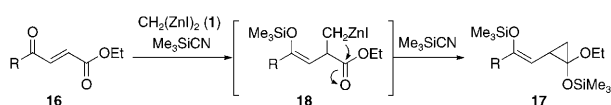
Scheme 5. Cyclopropanation of ketoester **16a** with bis(iodozincio)methane (**1**).

Table 2. Cyclopropanation of ketoesters **16** with bis(iodozincio)methane (**1**).

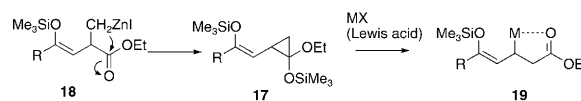
entry	substrate	R ₃ SiX	product	yield (%) ^[a]
1		Me ₃ SiCN		99
2		Me ₃ SiCN		73
3		Me ₃ SiCN		99
4		PhMe ₂ SiCl		83
5		PhMe ₂ SiCl		93
6		PhMe ₂ SiCl		90
7		PhMe ₂ SiCl		65
8		PhMe ₂ SiCl		97

[a] Isolated yields.

with Lewis acid generates homoenolates,^[8] which can be applied for the C–C bond forming reactions.^[9] When **17** was activated by a Lewis acid, the homoenolate **19** would be generated by cyclopropane ring opening (Scheme 7); the homoenolate **19** is considered to be a 3-siloxyallylmetal species.^[10] This reagent might have an allylmetal form **19**, because of the coordination between the metal and carbonyl group. The silyl enol ether was formed as the (*Z*)-form selectively. So, a nucleophilic reaction of **19** as 3-siloxyallylme-



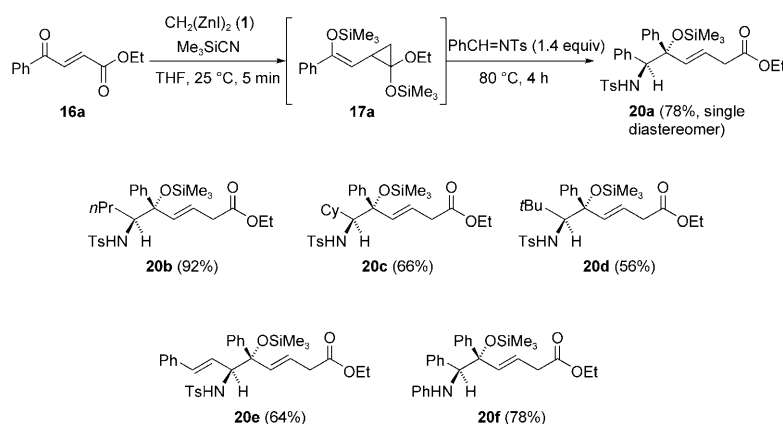
Scheme 6. Formation of cyclopropane **17** by 1,4-addition.



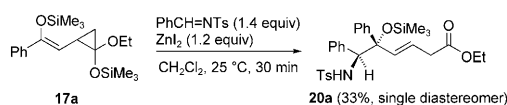
Scheme 7. Strategy of migration of iodozincio group from homoallyl metal into allyl metal.

tal reagent was expected to proceed regio- and diastereoselectively.^[11]

Treatment of the tosylimine of benzaldehyde with **17a** in the presence of zinc(II) iodide gave the β -siloxyamine derivative **20a** as a single diastereomer (Scheme 8). The stereochemistry of **20a** was determined by X-ray analysis (Figure 2).^[12] Meanwhile, the same reaction from **17a** without zinc(II) iodide resulted in a complete recovery.



Scheme 9. Reaction of various imines with the homoenolate prepared from **16a**.



Scheme 8. The reaction of **17a** as homoenolate with imine.

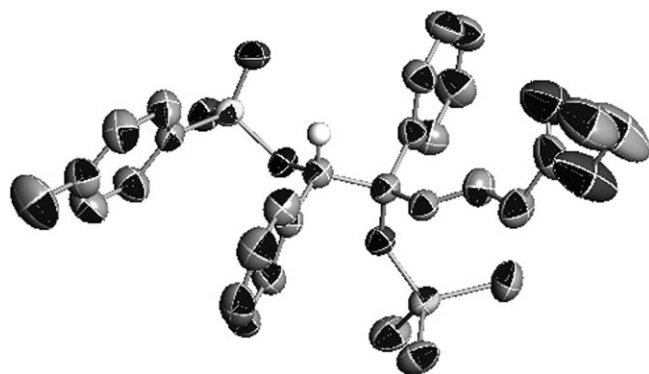


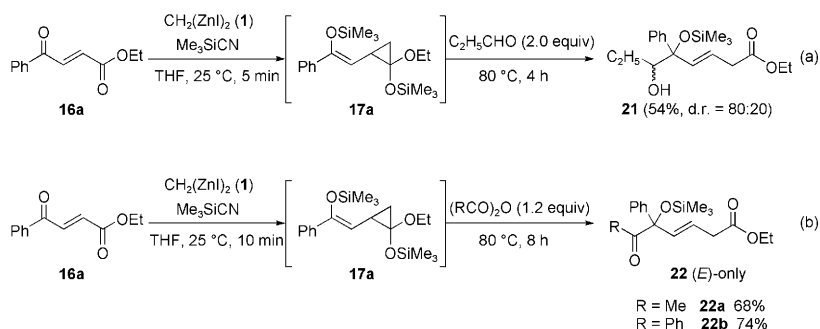
Figure 2. The ORTEP of **20a**.

When the ketoester **16a** was treated with bis(iodozincio)-methane (**1**) in the presence of trimethylsilyl cyanide as in Scheme 5, zinc(II) salts such as zinc(II) iodide and cyanide were also generated in situ. These zinc salts might mediate the conversion of the 1-trimethylsiloxy-1-ethoxycyclopropane moiety in **17a** into the zinc homoenolate. So we tried the cyclopropanation reaction and the addition reaction in one pot without the isolation of silyl ether **17a**. For this purpose, after the procedure for the preparation of **17a** from **16a** without addition of triethylamine, we added the tosylimines and then raised the reaction temperature to 80 °C for

4 h. As shown in Scheme 9, β -siloxyamine derivatives **20** were obtained with high diastereoselectivity. Instead of tosylimine, treatment of the phenylimine of benzaldehyde with **17a** also gave the corresponding β -siloxyamine derivative **20f** diastereoselectively.

Instead of imines, other electrophiles were also examined. Treatment of in situ formed **17a** with propanal under the reflux condition afforded the *vic*-diol **21** in moderate yield. The product **21**, however, was given as a diastereo mixture (Scheme 10a). The reaction of acid anhydrides with **17a** afforded α -(*E*)-alkenyl- α -siloxyketones **22** in good yields with high (*E*)-selectivity (Scheme 10b).

We propose the explanation for the high diastereoselectivities in the reaction of imines with the homoenolate, as shown in Figure 3. The conformation of the 3-siloxyallylzinc intermediate might be fixed by the chelation between a zinc atom and two oxygen atoms as (*Z*)-**23a**. The nucleophilic reaction proceeds regioselectively, retaining the conformation, and as a result, the product is afforded as the (*E*)-form dominantly. Furthermore, considering the trajectory of nucleophile to imine, we propose the reaction pathway shown as **24**. In comparison with another possible intermediate **24'**, **24** is much more favourable, since the intermediate **24'** contains a steric interaction between an alkyl group on the imine and a phenyl group in the reagent. In addition, an intermolecular chelation between a siloxy group and nitrogen atom of



Scheme 10. Reaction of propanal or acid anhydride with the homoenolate from **16a**.

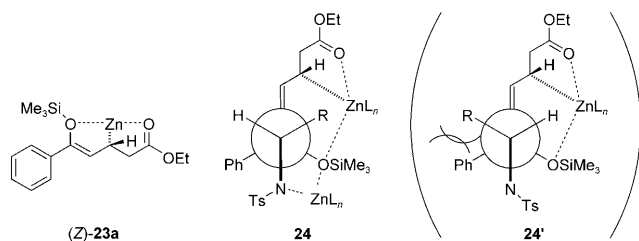


Figure 3. Proposed conformation of (Z)-23 and the transition state 24.

the imine would benefit the intermediate **24**. So, the nucleophilic attack of (Z)-**23a** to an imine proceeds via **24** to generate a couple of stereogenic centers diastereoselectively, and as a result, the *threo* isomer **20a** will be a major product.

Conclusions

We have demonstrated the nucleophilic cyclopropanation reaction of γ -acyloxy- α,β -unsaturated ketone or γ -ethoxycarbonyl- α,β -unsaturated ketone with bis(iodozincio)methane in the presence of silylating reagent. The reaction proceeded by 1,4-addition and sequential intramolecular nucleophilic attack to adjacent electrophilic sites to afford cyclopropane derivatives or 1-ethoxy-1-siloxy-1-cyclopropane derivatives in good yield. Furthermore, we also demonstrated the reaction with imines of (Z)-3-siloxyallylzinc species, which can be prepared easily from the obtained 1-ethoxy-1-siloxy-1-cyclopropane derivative with the aid of zinc salts existing in the reaction mixture. The reaction proceeded with high diastereoselectivity to give 1-(*E*)-alkenyl-2-(1-aminoalkyl)alkanol.

Experimental Section

Bis(iodozincio)methane (1): A mixture of Zn (25 mmol), diiodomethane (1.0 mmol), and PbCl_2 (0.01 mmol) in THF (2.0 mL) was sonicated for 1 h in an ultrasonic cleaner bath under Ar. To the mixture, diiodomethane (10 mmol) in THF (20 mL) was added dropwise over 15 min at 0°C with vigorous stirring. The mixture was stirred for 2 h at 0°C. After the stirring was stopped, the reaction vessel was allowed to stand undisturbed for several hours. Excess zinc was separated by sedimentation. ^1H NMR spectra of the obtained supernatant showed a broad singlet at -1.2 ppm at 0°C, which corresponded to the methylene proton of **1**. The supernatant was used for the further reaction as a solution of **1** in THF (0.5–0.6 M). Bis(iodozincio)methane in THF can be kept unchanged at least for a month in a sealed reaction vessel.

General procedure for the nucleophilic cyclopropanation of γ -acyloxy- α,β -unsaturated ketone: A mixture of γ -acyloxy- α,β -unsaturated ketone (**11**, 1 mmol), triethylamine (2.5 mmol), and trimethylsilyl cyanide (2.2 mmol) in THF (4 mL) was treated with bis(iodozincio)methane (**1**, 0.5 M solution in THF, 1.2 mmol) at 25°C for 3 h. Then, water was added to quench the reaction and extracted with hexane. The combined organic layers were washed with brine, and dried over Na_2SO_4 . The solvent was removed under reduced pressure, and the residue was purified by silica gel column chromatography to give the corresponding cyclopropane **15**.

General procedure for the nucleophilic cyclopropanation of (*E*)-ethyl 4-oxoalk-2-enoate: A mixture of (*E*)-ethyl 4-oxoalk-2-enoate (**13**, 1 mmol), triethylamine (2.5 mmol), and trimethylsilyl cyanide (1.1 mmol) in THF

(4 mL) was treated with bis(iodozincio)methane (**1**, 0.5 M solution in THF, 1.2 mmol) at 25°C for 5 min. Additional trimethylsilyl cyanide (1.1 mmol) was added and the mixture was stirred for 10 min at room temperature. Triethylamine was added and the whole mixture was poured into water, and the organic layer was extracted with hexane–EtOAc. The combined organic layers were washed with brine, and dried over Na_2SO_4 . The solvent was removed under reduced pressure, and the residue was purified by silica gel column chromatography to give the corresponding cyclopropane **17**.

General procedure for the tandem reaction of bis(iodozincio)methane (1**), **13a**, and imines:** A mixture of (*E*)-ethyl 4-oxoalk-2-enoate (**13**, 1 mmol) and trimethylsilyl cyanide (1.1 mmol) in THF (4 mL) was treated with bis(iodozincio)methane (**1**, 0.5 M solution in THF, 1.2 mmol) at 25°C for 5 min. Additional trimethylsilyl cyanide (1.1 mmol) was added and the mixture was stirred for 10 min at room temperature. A solution of imine (1.2 mmol) in THF (1.5 mL) was added, and the resulting mixture was heated to reflux for 4 h. The mixture was cooled to 0°C and poured into water to quench the reaction. The organic layer was extracted with EtOAc. The combined organic layers were washed with brine and dried over Na_2SO_4 . The solvent was removed under reduced pressure, and the residue was purified by silica gel column chromatography to give the corresponding (*E*)-1-alkenyl-2-(1-aminoalkyl)alkanol **20**.

Acknowledgements

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